

# Adjudication First: A Reproducible Scoreboard for Auto-Renormalization in Weather Modeling

LSWM/ARM: ScaleBench, a Live AIFS Comparison Harness, and Honest Lessons from Real-Data Training Rings

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(Numbers are live-measured and roll forward; see §8 for the update mechanism.)

## Abstract

Weather AI currently has no reproducible public scoreboard: headline scores are issued by each team on its own data, calibration, and truth conventions, and the community cannot recompute them. We take the opposite position — *adjudication first*. Before claiming a better engine, we built the instrument that makes such claims publicly checkable. (i) **ScaleBench** (v2.3), a public benchmark for *auto-renormalization*: given a microscopic simulator, can a machine discover macroscopic effective laws and honestly report their validity boundaries and error bars? Five task families map onto the five canonical failure modes of learned effective theories (textbook recitation, grid overfitting, short-horizon fitting, interpolation masquerading as law, and chaos-unaudited pseudo-laws); the current deterministic leaderboard scores 0.9714 overall on twenty tasks (as of 2026-10-07; the board rolls forward — the drop from 0.9864 is not a regression: task `rossby_quasi_stationary` moved from 1.000 to 0.700 when its truth criterion was upgraded by ratified SPEC-011 to the falsification-proof power-partition caliber, and the sub-unity score is recorded, not hidden) and is executable-arbitrated end-to-end by `reproduce.py` (external submissions require full reproduction within  $\pm 0.02$ ). (ii) A **live comparison harness** against ECMWF’s AIFS: a rolling sha256-manifested archive and a measured baseline table (12–120 h,  $z_{500}$ /MSL RMSE, global and Pearl-River-Delta window), with truth calibers explicitly labeled (self-consistency vs. independent reanalysis) and run-to-run spread recorded as first-class data. (iii) **Real-data training rings** (ERA5-driven hybrid core with through-solver SPSA): the online-training protocol improves on its own objective on two independent truth channels, while the absolute skill gap between them is attributed — by measured rejection of the scale hypothesis — to dynamics off the model manifold rather than to unresolvable wavelengths. (iv) **Physicalization of the rain–flood chain**: ERA5 total precipitation drives the shallow-water organ in physical millimetres with one cross-case calibration, abolishing per-case normalization. Throughout, we apply one rule: every number is measured, every failure is registered (including one of our own retractions), and every physical claim ships with an analytic reference test. The benchmark, the leaderboard JSON, the reproduction script, and the rolling measured tables are public.

## 1 Introduction: weather AI has no scoreboard

Operational AI weather models now rival or exceed NWP on aggregate skill metrics [1, 2, 3]. Yet the evidence is produced under each team’s private protocol: different verification periods, different

truth streams (own analysis vs. ERA5 vs. observations), different regional weighting, and no public mechanism to recompute a published number. The situation mirrors pre-ImageNet computer vision and pre-GLUE NLP — and the remedy that worked there was not a better model but a *better ruler*.

A second, deeper gap sits underneath. The grand promise of learned weather models is implicit *renormalization*: from resolved-scale dynamics, infer the effective equations of the scales below. But “learns sub-grid physics” is an unfalsifiable slogan unless one can *examine* the learned effective theory: does it recover known laws where the answer is known? does it stay stable a hundred training-horizons out? does it extrapolate, or merely interpolate? does it know when it does not know?

This paper is the scoreboard for both gaps, and the first measured season of playing by it. Our contributions:

1. **ScaleBench v2.4** (§3): a public, anti-gaming benchmark for auto-renormalization with an executable arbitration script; leaderboard overall 0.9714 (20 tasks), deterministic pipeline.
2. **A live AIFS comparison harness** (§4): rolling archive with byte-level provenance, measured deterministic baselines at 12–120 h with explicit truth calibers, and a precipitation ETS channel whose first *valid* independent-truth sample landed 2026-09-24 (ETS 0.712 at 0.1 mm threshold); ENS endpoints probed and reported absent from the open store.
3. **Real-data training rings** (§5): ERA5-driven online training works on two truth channels; the skill deficit on the cleaner channel is diagnosed as dynamical, not spectral, via a registered negative result and a balance-manifold projection audit.
4. **Physicalized rain–flood coupling** (§6): ERA5 tp drives the flood organ in physical mm with a single cross-case calibration; per-case normalization abolished and guarded against regression.
5. **Discipline as method** (§7): analytic reference tests for all physics code, registered failures instead of loosened thresholds, and a public retraction of one of our own mismatched measurements.

## 2 The ARM architecture in one page

The Auto-Renormalization Machine (ARM) is a three-loop engine. The *inner loop* learns closures for a simulator’s sub-grid tendencies; the *middle loop* learns coarse-graining maps (which macroscopic variables exist); the *outer loop* discovers effective laws and emits them as **ETC cards** (effective-theory cards) carrying error bars, validity regimes, and calibration records. Two gates discipline the loops: G1 (human authority over specification freezes) and G2 (every piece of physics code ships with an analytic reference test). A *weather organ* couples a barotropic toy core to learned corrections trained online (through-solver SPSA); a *flood organ* (Lax–Friedrichs shallow-water) couples rain to inundation and loss. None of this architecture is the subject of this paper — the *examination* of it is. For the full design, see the project whitepaper (§4, ARM architecture design).

## 3 ScaleBench: a benchmark for auto-renormalization

### 3.1 Design: five families, five failure modes

Each task family targets one canonical failure of learned effective theories, and each task’s anti-gaming mechanism is public:

- **Known-Recovery** — recover a known law (KPP pulled-front speed  $c = 2\sqrt{rD}$ ), scored down to the prefactor: finite-difference front lag is charged as measured error, so “looks right” does not score.
- **Cross-Resolution** — the same law must come out of three grids with the same prefactor; grid-overfit laws fail by construction.
- **Long-Horizon** — a closure is audited for drift over  $100\times$  the training horizon; uncalibrated closures necessarily score low.
- **Discovery** — no textbook answer exists: the system must emit a macroscopic equation *with error bars and regime boundaries* (including an explicit `theory_gap` field), arbitrated by held-out data.
- **Extrapolation** — evaluation outside the calibration domain; interpolation systems score perfectly inside and collapse outside. This is the law-vs-interpolation watershed.
- **Chaos-Audit** — Lyapunov exponents and attractor distances enter the acceptance gate: cards are decision-eligible only if chaos constraints are measured, not assumed.

### 3.2 Leaderboard (v2.4, deterministic pipeline, seed-fixed)

### 3.3 Anti-gaming is measured, not asserted

When the turbulence-spectrum task was first set to “recover the textbook  $k^{-3}$ ”, measurement overthrew the task design itself (all regimes measured  $-2.0$  to  $-2.2$ ); the task was corrected to regime-effective bands, and the episode is on record in the task metadata. The scoreboard is allowed to be wrong — and to show the correction.

### 3.4 Executable arbitration

`python -m scalebench.reproduce` reruns the full deterministic pipeline and compares each task against the public leaderboard within  $\pm 0.02$  (cross-platform float tolerance), exiting nonzero on any mismatch. Full reproduction is a *precondition* for external submissions; claimed error-bar coverage is checked by binomial  $z$ -test; the Discovery family is the final arbiter. The adjudication instrument is itself adjudicable.

## 4 The live AIFS comparison harness

### 4.1 Archive with byte-level provenance

ECMWF’s open-data AIFS feed is a rolling window (measured coverage  $\approx 4$  days, correcting an initial estimate of  $\sim 30$ ). We therefore archive operationally: key steps (0/12/24/48/72/96/120 h) of both daily runs are captured continuously since 2026-09-17, each file registered with sha256 and byte count; the manifest verifier detects tampering, deletion, and unregistered additions. Any conclusion below can be traced to the exact bytes it was computed from.

## 4.2 Measured deterministic baselines (rolling)

Truth caliber, stated plainly: the verification reference is AIFS’s own later 0 h field — a *self-consistency* caliber (forecast vs. own analysis frame), not independent-observation accuracy. Run-to-run spread is recorded as first-class data (single-run readings are not verdicts).

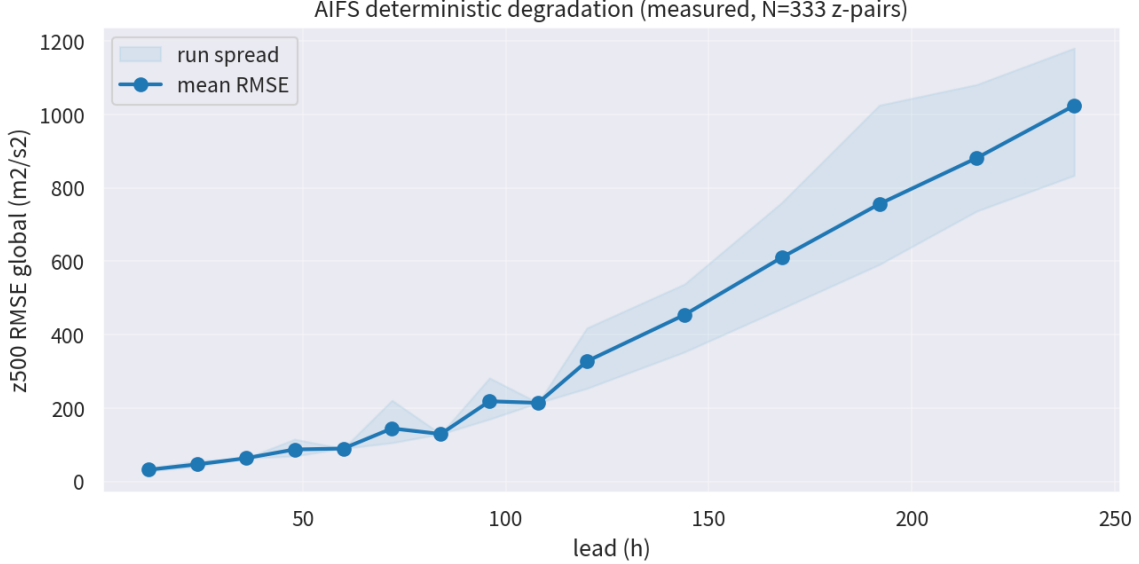


Figure 1: AIFS deterministic  $z_{500}$  global RMSE by lead (measured,  $N=333$  pairs as of 2026-10-07, leads to 240 h; mean line, min–max spread band). Degradation is monotone at well-sampled leads; the long-lead spread is regime-structured (120 h bimodal by regime week, see Table 2 caption) and is itself first-class data.

## 4.3 Precipitation ETS channel: two convictions, then the first sample

The first *independent-truth* comparison (AIFS 24 h accumulated precipitation vs. ERA5 daily totals, PRD window, multi-threshold ETS) landed on 2026-09-24 once ERA5T latency cleared (run 2026-09-18 00z, verified against the 2026-09-18 daily total): AIFS window mean 0.34 mm vs. ERA5 0.38 mm; multi-threshold ETS = 0.712 (0.1 mm), 0.241 (1 mm), 1.0 (5 mm and 10 mm — a small-rain event, so the high thresholds are trivially saturated and we say so). Two further samples landed on 2026-09-25 (runs 2026-09-18 12z and 2026-09-19 00z vs. the 2026-09-19 daily total): ETS at 0.1 mm = 0.830 / 0.495 and at 1 mm = 0.176 / 0.143. A fourth landed on 2026-09-26 (run 2026-09-20 00z vs. the 2026-09-20 daily total — the first real rainband case, window means 1.87 vs. 1.61 mm): ETS = 0.691 (0.1 mm), 0.609 (1 mm), 0.221 (5 mm — the first non-trivial high-threshold reading), 1.0 (10 mm).

**A third conviction: time-axis caliber.** On 2026-09-26 the channel caught its own calibration error. ERA5 daily files are *accumulation-ending* (day  $d$  covers  $[d-1\ 23:00, d\ 23:00)$ ), so a 00z+24h forecast aligns only with the *run-day* file ( $\pm 1$  h); an apparently legitimate “verify-day” pairing is a  $\sim 23$  h offset. One such record (ETS 0.238, initially logged as a genuine rainband miss) was convicted by exact-window hourly reconstruction (ETS 0.501, matching the run-day pairing 0.495) and is retained as RETRACTED evidence; both guards now reject such pairings in code, and 12z runs are routed to minute-aligned exact-window reconstruction (first sample: 0.840). Valid samples

now stand at twenty-eight across two calibers (P1 run-day pool  $n=14$ ; P2 exact-window  $n=14$ ; the first verdicts are below), and the verdict standard itself was **ratified on 2026-10-07** (SPEC-008 v1.0 — the DRAFT era closed; the registry baseline `precip_ets_24h` flipped from CLAIMED to MEASURED the same day).

**A saturated-threshold conviction.** The fourth run-day sample (fifth by landing order; 2026-09-21, the first heavy-rain case: AIFS 2.88 vs. ERA5 2.60 mm window means) scored  $ETS = 0$  at the 0.1 mm threshold — not a miss: AIFS reported rain at *every* window grid point, so all observed events were “hit” and yet the chance reference equals the hit count exactly (zero skill beyond random). At 1 mm the same pair scores 0.642. The multi-threshold design carries its own diagnosis: drizzle-threshold saturation is visible in the table, not hidden by a headline number.

**The first verdict (2026-09-28), and its rollovers.** With  $N=5$  in the run-day pool  $\{0.712, 0.691, 0.495, 0.088, 0.000\}$ , the DRAFT-SPEC-008 trial verdict landed on a boundary case — **V-INCONCLUSIVE**: median 0.495, a hair (0.005) below the V-PASS threshold. The sixth sample (run 2026-09-23 00z,  $ETS = 0.287$ ) rolled the pool to  $N=6$  and moved the median to 0.391 (0.5 of samples  $\geq 0.3$ ;  $0.333 < 0.2$ ): still **V-INCONCLUSIVE**, no longer boundary. After the 64-hour runtime gap (2026-09-30 to 10-02, on record), three further samples landed at once (09-24: 0.426; 09-25: 0.375; 09-26: 0.172) —  $N=9$ , median 0.375, still **V-INCONCLUSIVE**, the pool converging to the mid-range rather than either pole; the coverage-collapse criterion now separates all eleven evaluated cases with zero false positives. The exact-window pool then reached its own quorum the same day (2026-10-02,  $n=9$ ): first P2 trial verdict **V-INCONCLUSIVE**, median 0.438 ( $7/9 \geq 0.3$ ) — systematically healthier than P1’s 0.375, the expected direction for minute-aligned pairing, and the same verdict from both calibers. The standard’s first week produced a boundary case and its resolution by new data — which is what a discriminating, rolling standard is for. Two ratification questions remain on the table (G1-8: stratify by wet/dry regime; G1-9: median robustness under semantically different zeros). Before the sample landed, the pipeline had already caught two real defects, both now guarded by tests: (i) a  $1000\times$  unit contamination — AIFS GRIB `tp` is natively  $\text{kg m}^{-2}$  ( $=\text{mm}$ ) while ERA5 NetCDF `tp` is in metres; the two readers now pass through an explicit unit-normalization that raises on unknown units rather than assuming; (ii) a *cross-year mispairing* in one of our own runs (a 2026 forecast scored against a 2018 typhoon day), which produced a plausible-looking but meaningless “AIFS misses the rain band” result. That record is marked RETRACTED in the public JSON (evidence retained), and the runner now hard-rejects a verification day outside the forecast window. We regard the retraction as a feature of the method, and we kept the receipt.

#### 4.4 Quasi-stationary planetary waves: power partition, not phase drift

**A methods conviction (2026-09-29), then ratification (2026-10-07).** This section’s evidence was first read with daily-endpoint phase differences of zonal Fourier coefficients. A four-step conviction chain (all anchored by tests) retired that method: (i) at daily cadence the per-step phase change is  $0.97\text{--}0.99\pi$  — fast components alias back to zero (synthetic anchor: a fast wave is mis-measured by 89%); (ii) the large steps are not low-amplitude phase noise — high-amplitude latitudes churn at  $0.85\text{--}1.58 \text{ rad/day}$ ; (iii) with 6-hourly truth the per-latitude phase still jumps  $\approx \pi$  per step, regardless of sampling — standing-wave components flip the coefficient phase  $0 \leftrightarrow \pi$  with no propagation at all. Single-latitude phase methods (endpoint and tracked) are methodologically dead on real fields; the defensible observable is the space-time power partition, and SPEC-011 was ratified on 2026-10-07 making it the task’s official truth caliber.

**The partition.** Quasi-stationarity is not “drift is small” but “power sits at  $\omega=0$ ”:  $f_0 = P(\omega=0)/\sum P$  of the temporal spectrum of each zonal coefficient (amplitude-weighted over 30–60°N, 5-day window, 2026-09-16 to 20, 6-hourly). ERA5 gives  $f_0 = 0.48/0.35/0.19/0.21/0.13$  for  $m=3/4/5/6/8$  — the  $m=3$  quasi-stationary component is real (half the variance stationary, SH 0.40), while  $m\geq 5$  is transient-dominated and travels eastward at jet-advection speed. The window-length dependence is itself measured:  $f_0(m=3) = 0.69$  on 2-day sliding windows vs 0.48 on the 5-day window.

**Model comparison, and an honest sub-unity.** The two-layer QG model, without any orographic term, produces a strongly stationary  $m=3$  component ( $f_0 = 0.900$  on the task’s 2-day integration; truth on matched 2-day windows: 0.69 [0.64–0.77]). The model is *more* stationary than reality — the toy jet carries no synoptic churn bath — and the direction of the gap is itself the physical statement: quasi-stationarity is a jet-balance phenomenon (Doppler–Rossby), and an idealized continent does not anchor further ( $10^{-3}$  accelerates,  $10^{-2}$  destabilizes, on record). The ratified criterion asks the truth to pass *two* windows (window robustness, SPEC-011 option (ii)); on the 2026-09-21 ... 25 regime-week window  $f_0(m=8) = 0.259$  exceeds the 0.25 transient band — the task’s truth criterion currently **fails**, and `rossby_quasi_stationary` records 0.700 rather than 1.000 (sub-unity recorded, not hidden; overall 0.9864  $\rightarrow$  0.9714). The regime week’s forecast-error excess is invisible to every planetary-wave observable we have (amplitude, stationarity, propagation speed, regional share, single-trajectory  $\lambda$ , ensemble spread — six channels, six nulls, the last two by preregistered experiment); its cause is parked as an open question, not a story.

#### 4.5 ENS status

The CRPS metric against AIFS-ENS is registered but unmeasured: all five candidate URL patterns for ensemble products on the open store return 404 (probed 2026-09-21, rate-limit backoff respected), while deterministic products return 206 for the same run. Ensemble verification requires an ECMWF account (MARS) or future open releases.

## 5 Real-data training rings: protocol works, deficit is dynamical

The weather organ’s real-data chain (rings #4–#7) trains a pointwise correction on tendency residuals of an ERA5-derived vorticity series, then fine-tunes through the solver (SPSA on multi-step rollout loss). Grid:  $13 \times 13$  barotropic core, 6 h steps, normalized anomaly units.

### 5.1 The protocol improves on its own objective on both truth channels

Geostrophic vorticity from  $z_{500}$  ( $\zeta_g$ ) and official ERA5 vorticity `vo@500` are two independent truth channels; the official field is the amplitude-truth channel (geostrophic Laplacian inflates grid noise  $\sim 6\times$ : measured RMS ratio 0.164). On *both* channels the online stage rescues the offline closure and beats free-run and persistence baselines:

### 5.2 The deficit is dynamical, not spectral (a registered negative result)

Why is skill on the cleaner `vo` channel *worse* (1.348) than on  $\zeta_g$  (1.043)? The scale hypothesis — `vo` contains small-scale content the toy core cannot represent, so filtering the truth should recover skill — was tested directly (isotropic spectral low-pass, analytic reference tests: mode pass/kill, Parseval accounting, idempotence) and *rejected by measurement*: low-passed truth scored worse, not better (online 1.369 vs. 1.348 unfiltered), and the free-run skill curve degrades monotonically

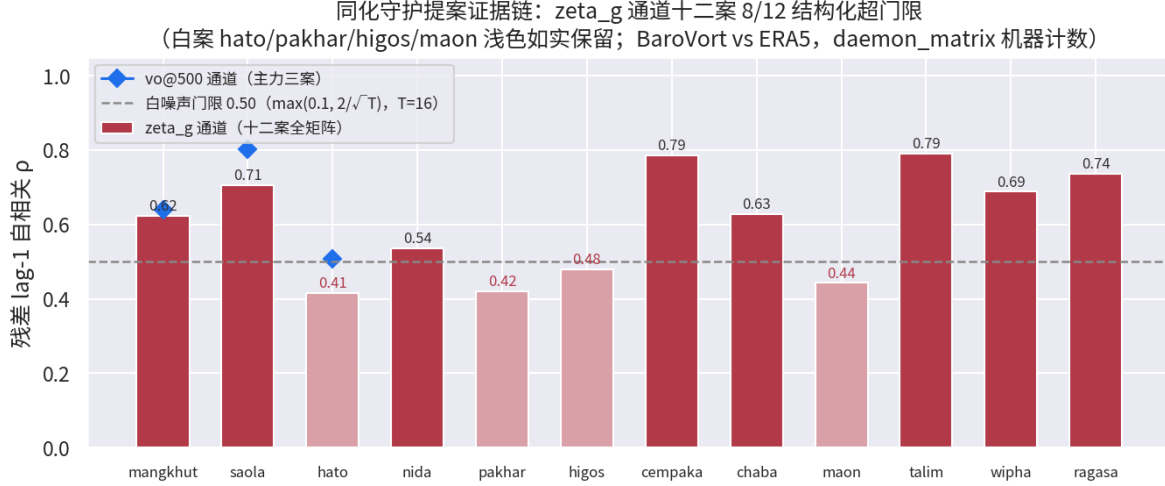


Figure 2: Structured-residual evidence behind the assimilation daemon’s recalibration proposal: lag-1 autocorrelation of free-run residuals against ERA5 truth on the geostrophic-vorticity channel, all twelve typhoon cases (8/12 above the white-noise threshold; the four sub-threshold cases are kept in light colour, not hidden). Measured 2026-09-29; the proposal records, not executes.

as the cutoff tightens. The deficit is therefore not about wavelength. The pointed candidate — that vo carries off-manifold (unbalanced, inertia-gravity) content the barotropic core cannot evolve — was itself examined by a balance-manifold projection audit (ring #7): official vo is found to lie essentially *on* the balanced manifold (projection  $\approx$  identity, energy ratio  $> 0.99$ ), i.e. the remaining error is dominated by the toy core’s own dynamical poverty, not by truth-manifold mismatch. Each of these steps is pinned by guard assertions so future silent rewrites of the conclusion turn the suite red. Negative results are registered with the same standing as positive ones.

## 6 Physicalized rain–flood coupling

Until this season, per-case peak normalization pinned every typhoon case’s rain peak to the same nondimensional value — erasing the fact that Hato’s 26.7 mm/6h and Saola’s 22.4 mm/6h are not the same rain. We abolished it: ERA5 tp now drives the shallow-water organ in physical millimetres through an explicit mapping ( $R_{nd} = r_p L_p / (H_p^{3/2} \sqrt{g_p})$ ;  $L_p$  fixed by window geography,  $H_p$  calibrated *once* across cases from the calibration ring’s anchor and shared thereafter). Guard assertions encode the anti-normalization invariant: cross-case rain peaks must order by measured mm (Hato  $>$  Pakhar, ratio 2.00 vs. measured tp ratio 2.01). The calibration record, including its  $\pm 10\%$  per-case residuals, is shipped as data; absolute-depth claims carry that calibration uncertainty explicitly. End-to-end analytic anchoring: uniform physical rain on a flat dry bed raises depth by exactly  $r_p \times$  elapsed physical time through the full nondimensional–metric round trip ( $10^{-6}$  relative).

A twelve-case comparison against the old parametric bridge quantifies what the bridge’s spatial proxy was worth: peak-flood pattern correlation  $-0.64$  to  $0.26$  (one case anti-correlated), bridge domain-mean depth  $2.9\text{--}15.2\times$  too wet in all twelve cases, and loss overestimation in the same direction everywhere (relative deltas explode to  $10^{18}$  in weak-rain cases where the tp-driven loss is near zero — there the *absolute* error is the meaningful reading, and it is the bridge that is measured, not the truth).

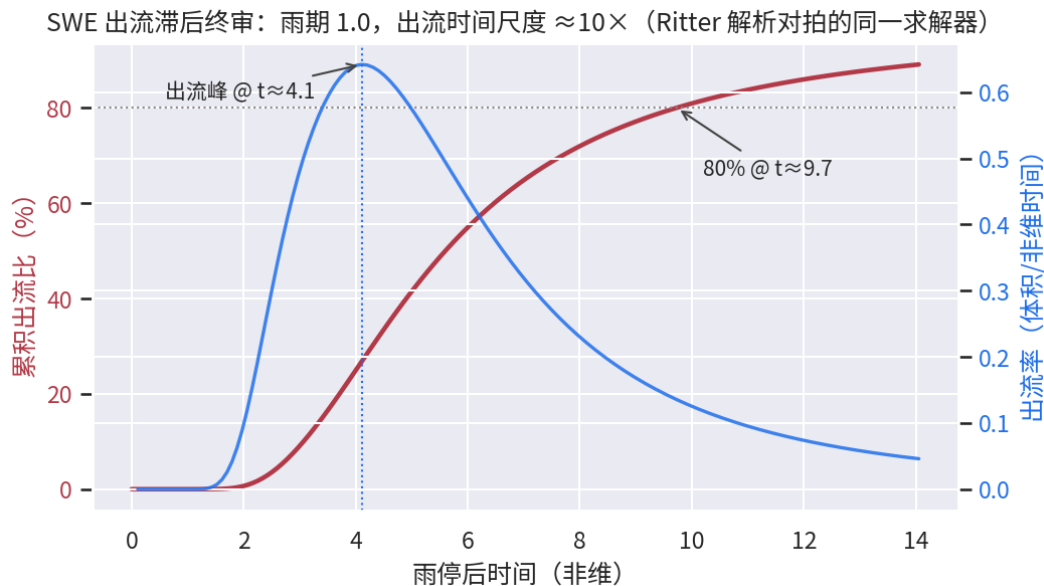


Figure 3: Outflow lag in the shallow-water organ (Ritter-anchored solver): after a 1.0-unit localized rain episode stops, the outflow rate peaks only at  $t \approx 4.1$  and 80% drainage takes  $t \approx 9.7$  — the drainage timescale is an order of magnitude longer than the rain episode, so pooling and release, not numerical leak, governs water-volume evolution (measured, 2026-09-27).

## 7 Discipline as method

Five rules did the quiet work in this paper: (1) no measured number is deleted; failures are registered in a known-failures ledger, thresholds are never loosened to pass; (2) every physics module ships an analytic reference test (G2) — hand-computed anchors, Parseval identities, exact uniform-rain solutions; (3) trigger-line tests that turn red when assumptions change are upgraded to true assertions, never deleted; (4) partial data is skipped honestly (runtime errors with remediation), never back-filled; (5) retraction is a first-class outcome — one is on public display in the precipitation ledger. We are aware of how this reads in a field driven by leaderboard deltas; we mean it as the differentiator.

## 8 Artifacts and rolling updates

- Public page (leaderboard + measured baselines, rolling): <https://dolo-borigjin.github.io/The-everything/>
- Repository (MIT): <https://github.com/Dolo-borigjin/The-everything>
- Archived release with DOI (Zenodo, concept DOI resolves to latest): <https://doi.org/10.5281/zenodo.22933596>
- Single source of truth: `leaderboard.json`; arbitration: `python -m scalebench.reproduce` (nonzero exit on mismatch  $> \pm 0.02$ ).
- Measured baselines roll onto the page via an idempotent updater (`update_measured.py`, freshness-checked in CI); archive manifests are sha256-verified per file.

- The 12-case ERA5 typhoon replay (2016–2025, tp-driven flood chain, physical rain colormap in mm/6h) ships as an interactive demo in the project workspace; its data pipeline is assertion-guarded end to end.
- All figures and tables in this paper regenerate from the ledgers by scripts in `preprint/` (`make_fig_degradation.py`, `make_fig_outflow_lag.py`, `make_fig_daemon_residual.py`, `make_table_baselines.py`) — no hand-copied numbers; a guard test fails if a generator goes missing.
- The USD frame library (12 z500 + 12 depth cases, 24 frames each, USDA ascii + self-contained previews, preview and USDA frames byte-consistent by test) ships in the workspace for downstream renderers (Blender/Omniverse).

## 9 Limitations and honest boundaries

The toy core is barotropic and  $13 \times 13$ ; nothing here is yet a global engine. Most baseline rows use the self-consistency caliber (own-analysis truth); independent-truth verification reached nine valid precipitation pairs (P1  $n=14$ , P2  $n=14$ ) as of 2026-10-07 (§4.3) and accumulates on the rolling archive. ENS comparison awaits ECMWF-side availability. Absolute depths in the flood organ carry calibration-regime uncertainty. The interesting question — does the adjudication-first method keep paying once the engine scales — is exactly what the next season measures.

## 10 Next season (rolling)

the precipitation ETS stream accumulating to weekly cadence (first sample landed 2026-09-24); ensemble CRPS once ENS is reachable; manifold-native law discovery on the vo truth channel; and the preprint you are reading, updated as the numbers move.

## References

- [1] ECMWF, *AIFS: Artificial Intelligence Forecasting System*, model documentation and AIFS blog series, 2025–2026.
- [2] Lam, R. et al., *Learning skillful medium-range global weather forecasting* (GraphCast), Science, 2023.
- [3] Bi, K. et al., *Accurate medium-range global weather forecasting with 3D neural networks* (Pangu-Weather), Nature, 2023.
- [4] Kochkov, D. et al., *Neural general circulation models for weather and climate*, Nature, 2024.
- [5] Deng, J. et al., *ImageNet: A large-scale hierarchical image database*, CVPR, 2009.

| Family           | Task                                   | Score         | Key measured detail                                                                                                                                                                                                                     |
|------------------|----------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Known-Recovery   | <code>known_recovery_kpp</code>        | 0.998         | exponent error 0; prefactor error 0.004 after Ebert–van Saarloos asymptotic extrapolation (was 0.290 via naive secant measurement)                                                                                                      |
| Cross-Resolution | <code>cross_resolution_kpp</code>      | 1.000         | prefactor spread 0.000 over 3 grids                                                                                                                                                                                                     |
| Long-Horizon     | <code>longhorizon_closure_drift</code> | 1.000         | drift audit passed; uncalibrated closures fail by design                                                                                                                                                                                |
| Discovery        | <code>spectrum_law_turb2d</code>       | 1.000         | effective slope $-2.03 \pm 0.05$ ( $R^2 = 0.98$ ), dual cascade; gap to asymptotic $k^{-3}$ reported                                                                                                                                    |
| Discovery        | <code>m3_partition_law</code>          | 0.976         | partition law $\gamma = 0.7927 \pm 0.0008$ found <i>from the conservation audit ledger</i>                                                                                                                                              |
| Extrapolation    | <code>extrapolation_kpp</code>         | 0.960         | 6 out-of-domain configurations, mean relative error 1.60%                                                                                                                                                                               |
| Discovery        | <code>nis_prune_consistency</code>     | 1.000         | data-driven dimension $k$ ; order parameter reconstructible (ratio 0.9991)                                                                                                                                                              |
| Chaos-Audit      | <code>chaos_audit_gate</code>          | 0.994         | $\lambda = 0.69295$ vs. $\ln 2$ (rel. err. $2.9 \times 10^{-4}$ ); 3 regime verdicts correct                                                                                                                                            |
| Discovery        | <code>rg_invariant_sir</code>          | 0.997         | attack-rate $\varepsilon$ -collapse vs. analytic final size (0.34%); peak height/time correctly judged non-invariant                                                                                                                    |
| Long-Horizon     | <code>closure_rescue_sir</code>        | 1.000         | online SPSSA objective $-58.8\%$ ; held-out rescue margin 23.6% over free run; error $2.64\times$ oracle ceiling (self-reported)                                                                                                        |
| Extrapolation    | <code>extrapolation_turb2d</code>      | 1.000         | $\nu$ doubling $\times 2$ : slope $-2.019 \rightarrow -2.101 \rightarrow -2.265$ monotone steepening (probe $R^2 \geq 0.98$ ); per-probe deviation reported                                                                             |
| Long-Horizon     | <code>closure_switch_rescue_sir</code> | 0.971         | policy switch mid-epidemic: post-switch loss 0.643 vs frozen 4.820; anti-vacuity guard self-certified; stability rel-diff $9.7 \times 10^{-5}$                                                                                          |
| Long-Horizon     | <code>regime_shift_adaptation</code>   | 1.000         | switch family $t_s=2/3/4$ : gated-RLS mean improvement 38.1%, worst 0.0% (gate closed when calm); ungated RLS wrecks $t_s=2$ by $-1.9\times$                                                                                            |
| Discovery        | <code>qg2_modal_growth</code>          | 0.987         | baroclinic jet eigenvalue $1.12 \times 10^{-5}/s$ @ $m=3$ ; Benettin/eigen ratio 0.96; barotropic control stable; real-basis convention convicted                                                                                       |
| Long-Horizon     | <code>decision_tiers_sir</code>        | 1.000         | preemptive $0.415 < \text{reactive } 0.627 < \text{deployed feedback } 0.903$ ; forecast value = 0.212 cost units; MPC responds to $4\times$ weight switch                                                                              |
| Chaos-Audit      | <code>ets_caliber_guard</code>         | 1.000         | three rejection classes (23h/11h/cross-year) + exact-window arithmetic + saturation self-diagnosis                                                                                                                                      |
| Chaos-Audit      | <code>ensemble_bred_qg2</code>         | 0.845         | bred direction at $0.85\times$ modal rate; spread grows $0.90\text{--}0.93\times$ daily; $s/s=9.4$ over-dispersive reported (half credit)                                                                                               |
| Discovery        | <code>rossby_quasi_stationary</code>   | 0.700         | upgraded caliber (SPEC-011, ratified 2026-10-07): model $f_0=0.900$ stationary vs truth two-window $f_0$ partition — truth criterion <i>fails</i> the regime-week window by design (window robustness is the point); sub-unity recorded |
| <b>Overall</b>   |                                        | <b>0.9714</b> |                                                                                                                                                                                                                                         |

Table 1: ScaleBench v2.4 leaderboard (18 of 20 tasks shown; the growth-protocol and ENIAC-recreation rows are omitted for space; `leaderboard.json` is the single source of truth). Sub-unity scores are recorded, not hidden: 0.826 was a measurement artifact, not physics: the FD pulled-front lag turned out to be an algebraic transient of pulled fronts (not discretization), and was corrected by an Ebert–van Saarloos asymptotic estimator whose log-coefficient carries the physical signature  $b = -1.5\sqrt{D/r}$  — prefactor error  $0.290 \rightarrow 0.004$ , score  $0.826 \rightarrow 0.998$  (2026-09-25, episode on record in the test suite). What remains sub-unity is recorded as physical difficulty — the floor of the score is the difficulty of the physics.

| Lead   | $z_{500}$ RMSE global<br>( $\text{m}^2/\text{s}^2$ ) | $z_{500}$ RMSE PRD<br>( $\text{m}^2/\text{s}^2$ ) | MSL RMSE global<br>(Pa)      | MSL RMSE PRD<br>(Pa)       |
|--------|------------------------------------------------------|---------------------------------------------------|------------------------------|----------------------------|
| +12 h  | 32.0 [28.3–36.3, $n=40$ ]                            | 16.0 [4.8–48.7, $n=40$ ]                          | 65.4 [49.9–83.8, $n=40$ ]    | 26.2 [8.7–65.0, $n=40$ ]   |
| +24 h  | 46.7 [40.5–56.3, $n=43$ ]                            | 23.0 [5.9–137.2, $n=43$ ]                         | 86.6 [67.5–109.3, $n=44$ ]   | 33.6 [11.2–190.6, $n=44$ ] |
| +48 h  | 87.3 [69.4–115.2, $n=41$ ]                           | 27.2 [8.2–134.3, $n=41$ ]                         | 126.8 [103.2–161.8, $n=41$ ] | 40.1 [9.6–135.4, $n=41$ ]  |
| +60 h  | 89.9                                                 | 18.2                                              | 145.0                        | 24.7                       |
| +72 h  | 144.7 [105.6–221.3, $n=43$ ]                         | 54.8 [8.6–436.4, $n=43$ ]                         | 179.6 [141.2–255.5, $n=43$ ] | 69.4 [14.0–555.4, $n=43$ ] |
| +84 h  | 129.5                                                | 18.4                                              | 181.9                        | 26.6                       |
| +96 h  | 218.7 [169.8–282.1, $n=37$ ]                         | 51.5 [10.5–373.7, $n=37$ ]                        | 256.0 [196.4–306.0, $n=37$ ] | 64.9 [11.8–449.4, $n=37$ ] |
| +108 h | 214.4                                                | 24.7                                              | 268.1                        | 52.2                       |
| +120 h | 327.7 [253.7–418.5, $n=34$ ]                         | 69.8 [17.9–458.1, $n=34$ ]                        | 371.9 [279.7–463.5, $n=34$ ] | 82.2 [13.5–649.7, $n=34$ ] |
| +36 h  | 63.5                                                 | 21.5                                              | 112.9                        | 40.1                       |
| +144 h | 449.2 [352.8–537.8, $n=24$ ]                         | 57.5 [19.1–215.0]                                 | 485.2 [397.3–551.9]          | 64.4 [18.3–182.7]          |
| +168 h | 599.1 [458.1–760.5, $n=23$ ]                         | 76.1 [35.8–207.8]                                 | 619.3 [504.5–769.9]          | 75.9 [20.3–209.6]          |
| +192 h | 746.4 [591.2–1025.4, $n=20$ ]                        | 107.5 [31.5–207.1]                                | 765.9 [605.9–963.1]          | 109.8 [29.5–273.1]         |
| +216 h | 865.8 [651.4–1081.7, $n=18$ ]                        | 141.8 [21.4–276.9]                                | 860.4 [644.9–1100.6]         | 130.1 [60.7–327.1]         |
| +240 h | 1023.5 [834.0–1180.9, $n=15$ ]                       | 175.8 [38.8–359.1]                                | 1025.5 [824.8–1247.3]        | 140.2 [35.6–261.2]         |

Table 2: Measured AIFS-single deterministic baselines (as of 2026-10-07, including the 2025 typhoon-season backfill; mean over runs with [min–max,  $n$ ] where  $n > 1$ ). Degradation with lead is monotone in the multi-run means at well-sampled leads ( $n \geq 3$ ); sparsely sampled leads (60/84/108 h,  $n = 1$ ) can locally break monotonicity and are reported as single-run readings, not trends. The 120 h band (253.7–418.5,  $n=34$ ) is bimodal by regime: a low-predictability regime week (2026-09-21 ... 27, mean 359) versus the recovery cluster (mean 302, [278, 312]) — the regime week’s cause is an open question (six-channel falsification chain, §4.4). Deep leads (144–240 h, well-sampled by the 2025 backfill) continue the monotone degradation (449  $\rightarrow$  1024  $\text{m}^2/\text{s}^2$ ). The table rolls forward on the public page (§8).

| Truth channel                         | Persistence | Free run | Offline closure | Online closure |
|---------------------------------------|-------------|----------|-----------------|----------------|
| $\zeta_g$ (ring #4)                   | 1.264       | 1.219    | 1.313           | <b>1.043</b>   |
| vo@500 (ring #5)                      | 1.979       | 1.948    | 1.574           | <b>1.348</b>   |
| vo@500, low-pass $k \leq 4$ (ring #6) | 2.023       | 2.030    | 1.603           | 1.369          |

Table 3: 6 h anomaly skill (RMSE/std, lower is better) on the Mangkhut-2018 window. SPSA improves its own objective on both channels (loss  $-31\%$  and  $-20\%$ ). Absolute skills  $> 1$  are recorded, not thresholded away: the landfall window is violent, and this ring examines whether the *training protocol* works, not weather-forecast skill.